# Aniket Navin Arbatti

Toronto, ON | Canadian Citizen | 647-537-9870 | Email | LinkedIn | GitHub

## SUMMARY

Third-year Computer Engineering student at Toronto Metropolitan University with experience developing **high-performance C++** software in **Linux** and **C/C++ firmware for embedded systems**. Skilled in performance profiling, cache and memory optimization, computer architecture and hardware/software integration. Seeking software engineering co-ops beginning May 2027.

## TECHNICAL SKILLS

- Languages: C/C++, **Java**, Python, Linux shell script
- Systems and Performance: Linux, Linux perf, CPU performance profiling, **memory management**, **multithreading**, **computer architecture, Data Structures & Algorithms**
- Concepts: **Object-Oriented Design**, Unified Modelling Language (**UML**), **Software Testing**
- Tools: **Git, GitHub**, Altium Designer

## EDUCATION

### Toronto Metropolitan University

Honours B.Eng. in Computer Engineering (Co-op) *SEP 2024 – JUN 2028*

- **cGPA: 3.97/4.33** | *Dean's List Scholar:* 2024, 2025, 2026
- Planned Coursework: Operating Systems, Computer Organization, Microprocessor Systems

## PROJECTS

### High-Performance Analysis of Binaro | GitHub *AUG 2026 – PRESENT*

- Built a **high-performance C++ puzzle solver** on **Linux** and benchmarked performance across 100,000 puzzles, establishing a baseline solve time of **30 µs/puzzle**
- Profiled CPU performance with **Linux perf**, analyzing cycles, instructions, cache misses and branch behavior to identify microarchitectural bottlenecks
- Replaced a branch-heavy and memory-intensive grid representation with a compact bitmask-based design, **reducing memory footprint by 80%** and **L1 data-cache misses by 37%**
- **Reduced branch mispredictions by 88%** and total CPU cycles by **55%,** improving solver runtime from **30 µs/puzzle to 8 µs/puzzle**
- Controlled CPU benchmark conditions to produce reproducible performance measurements with **<1% run-to-run timing variation**
- Developing a **multithreaded** solver architecture to parallelize puzzle processing across CPU cores and evaluate scalability

### Try Go! | GitHub *JUL 2026 – PRESENT*

- Developed Go! as a passion project after identifying a gap in existing solutions for collaboratively deciding what to do with friends
- Designed and implemented a full-stack collaborative workflow spanning activity discovery, persistence, group creation, location sharing, and voting using **Cursor** to accelerate development
- Built the application with **React, JavaScript, TypeScript, Vite, Tailwind CSS, Zustand, React Router, and Leaflet**, implementing a lightweight TypeScript API layer with custom Vite server plugins and deploying to **Vercel**

## WORK EXPERIENCE

### Avionics PCB/CAD Lead | MetRocketry | Toronto, ON | 2025/2026 PCB Showcase *OCT 2024 – PRESENT*

- Developed **firmware in C/C++** for sensor peripherals using **I2C**, **UART** and **SPI** on a **Teensy 4.0 Arm Cortex M7**, transmitting **7 flight telemetry parameters** at **1 Hz** for real-time monitoring and post-flight analysis
- Designed three custom two-layer PCBs in **Altium Designer** to serve as the rocket's telemetry flight computer
- Validated and **debugged** the system through **three integration tests** across four subteams and a successful **IREC** flight, maintaining telemetry from launch to recovery while operating under **extreme desert conditions** and **18 g launch acceleration**
- **Promoted** from **Avionics PCB/CAD Member to Lead** following 2026 competition performance

### Academic Assistant for the Course Electric Networks | TMU | Toronto, ON *JAN 2026 – AUG 2026*

- Evaluated low-cost instrumentation options for undergraduate analog circuit labs serving **200+ students per semester**, with the goal of reducing laboratory equipment requirements and enabling at-home experimentation
- Researched and tested the **FNIRSI 2C53P**, a 3-in-1 oscilloscope, function generator, and multimeter, as a potential alternative
- Developed procedures for op-amps, second-order circuits, filters, and transformers and prepared the necessary documents in **Microsoft Office** for **instructor review**